CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE)

May 2014

CE402 Advanced Operating System

Date: 23.05.2014, Friday **Time: 01:30 to 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Compare the IPC functionality provided by pipes and shared memory. What are the advantages and drawbacks of each. [04]

(b) Differentiate between process and thread. [03]

Q - 2 (a) Why context switch between two threads in the same process is cheaper than context switch between two threads in different process? Give valid reasons. [06]

(b) What do you mean by daemon? Differentiate background process and foreground process with an example. [04]

(c) Explain monolithic kernel with proper diagram. [04]

OR

(b) Give the differences: [06]

(1) User mode Vs. Kernel mode

(2) Distributed operating system Vs. Network operating system

(c) How to achieve synchronization in distributed operating system? [02]

Q - 3 (a) Explain ricart-agrawala algorithm with an example to achieve mutual exclusion in distributed system. [06]

(b) Explain in detail about PCB and how process represented in system memory. [05]

(c) Differentiate process context and system context. [03]

OR

(b) Explain ring-based algorithm to achieve mutual exclusion in distributed systems. [05]

(c) What are the design issues for thread package? [03]

SECTION – II

Q - 4 (a) Explain Layers, interfaces, and protocols in the OSI model. [04]

(b) How a conventional (i.e., single machine) procedure call works? Explain with an example. [03]

Q - 5 (a) What are the main differences between a LAN and a WAN? Why are medium access control protocol needed?. [06]

(b) List Various strategies are used to handle deadlocks in distributed systems. Explain one of strategy in brief. [05]

(c) Compare three process with one thread each and one process with three threads. [03]

OR

Q - 5 (a) Explain four algorithms for managing a client file cache. [05]

(b) What are the Implementation issues for Processor allocation algorithms in distributed systems. [05]

(c) Describe the differences between symmetric and asymmetric multiprocessing. [04]

Q - 6 (a) **Answer the following (Any Three)** [12]

(1) Draw and explain NFS layer structure.

(2) Explain multiprocessor with caching and without caching.

(3) What do you mean by strict consistency in distributed shared memory? Explain in detail.

(4) Explain fault tolerance using active replication.

(b) Differentiate named pipe and unnamed pipe. [02]

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